



**Department of Information Technology
Academic Year 2024 – 2025 (Odd Semester)**

Degree, Semester & Branch : III Semester-B. Tech Information Technology

Course Code & Title : CS3391 Object Oriented Programming

Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic : Unit 2 / Inheritance -Basics and Types of Inheritance**
- **Course Outcome : CO2**
- **Topic Learning Outcome: TLO6**
- **Activity Chosen : Mind Map**
- **Justification:**
 - A mind map is a visual representation of concepts and ideas. It is a visual thinking tool that aids to write down ideas quickly.
 - Inheritance is one of the object-oriented programming principles in which one object acquires the properties and behaviours of another object. It provides code reusability and establishes relationship between different classes.
 - In order to have a recap of inheritance and its types, this activity is chosen.
- **Time Allotted for the Activity: 15 Minutes**
- **Details of the Implementation:**
 - Faculty explained the basics of inheritance and its types with example program in the class room.
 - Based on the discussion and after clarifying the student's doubts, the faculty instructed the students to create a mind map related to the topic in 15 minutes.
 - Each Student created a mind map based on their level of understanding. The sheets were collected from the students by the instructor.
 - This activity helped the students to recall the topic taught that day, generate new ideas about the topic and to answer the questions easily as shown in Fig 2.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	3	2	2	1	1	3

(1 – Low 2 – Moderate 3 – High)



• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	Students will be able to understand the concept of inheritance and apply to real world problems	Students will be able to write and develop java program using inheritance	Students will be able to illustrate the concept of inheritance and its types as a mind map	As the student involved in the activity as an individual, the individual thinking skill improved	Students will be able to communicate effectively on complex engineering activities	The recollecting and thinking skill gained through this activity helps the students in solving real world problems

• Images / Screenshot of the practice:



Fig 1: Mind Map Activity

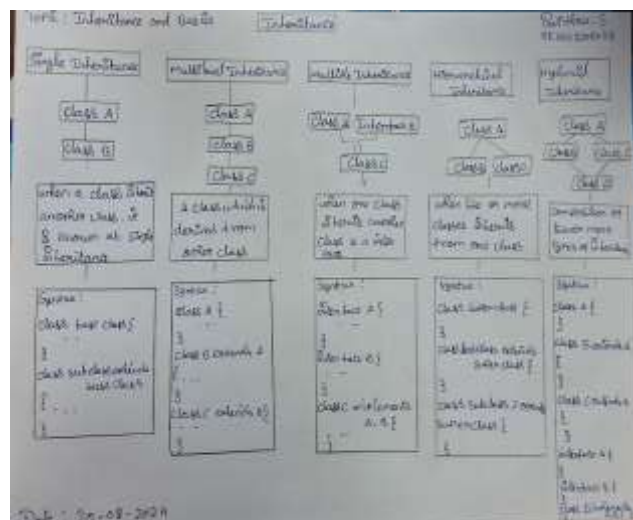


Fig 2: Sample Sheet of Mind Map Activity of Student S. Pavithra

• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Students said that the activity assisted them in determining their level of understanding of the concept.
- They said it would be useful to recollect the concept in short time.



- They felt comfortable to ask doubts with their peers.
- Students feel easy to understand the concept and to write programs.

❖ ***Benefit of the practice:***

- This activity enables students to recall the most important information using keywords, and then visually connect facts and ideas.
- The Mind Map activity made the slow learners to remember the concept easily.
- Students spent their time in self-learning.
- This activity encouraged the students to share their knowledge with others.
- From this activity, Students can get more clarity in the particular topic

❖ ***Challenges faced in implementation:***

- Some of the students represent very less key points in the mind map made a mistake in writing programs using access specifiers
- It took more time for some students to complete the activity.

The Student S.Pavithra represented the Mind Map of inheritance concept with following mistakes

- i) She presented the inheritance concept in chart with less key information
- ii) She didn't represent the extends keyword in the mind map diagram.
- iii) She made a syntax error in the multiple inheritance type

In the next class, I addressed the students how to connect all the key information of inheritance and its types in visual representation and also informed the students to present the mind map neatly.

References:

- https://www.ritrjpm.ac.in/images/computer-science/5_CS8591_Mindmap.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/Unit_5_Learning_by_Teaching.pdf

Signature of Faculty Member

HOD